# PASSWORD-PROTECTED DOOR LOCK USING STM32 MICROCONTROLLER

## A PROJECT REPORT ON

## PBCST 504 MICROCONTROLLERS

Submitted by

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**to**

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in partial fulfillment of the requirements for the award of the Degree

of

Bachelor of Technology

In

*Computer Science and Engineering*



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# DECLARATION

We undersigned declare that the project report **PASSWORD-PROTECTED DOOR LOCK USING STM32 MICROCONTROLLER**, submitted for the partial fulfillment of the requirements for the PBCST 504 Microcontroller subject in fifth semester under the university of the APJ Abdul Kalam Technological University, done by us under the supervision of the **Ms. MANUMEGHA M M, Assistant Professor CSE Department**. This submission represents our ideas in our own words and where ideas or words of others have been included. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other university.

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## CERTIFICATE

This is to certify that the report entitled "PASSWORD-PROTECTED DOOR LOCK USING STM32 MICROCONTROLLER ",submitted by ANAMIKA P(KMW24CS004) to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of **PBCST 504 Microcontroller** in fifth semseter in Computer Science and Engineering is a bonafide of project carried by them under our guidance and supervision. This report in any form not been submitted to any other University or Institute of any purpose.

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We would like to thank the almighty's mercy towards me over the years; it goes without saying that without them, I would not be where I am today.

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# ABSTRACT

Security systems play an important role in protecting homes, offices, and other restricted areas from unauthorized access. Traditional locking systems rely on physical keys, which may be lost, stolen, or duplicated.An automatic password-based door lock system provides an effective solution by allowing access only to authorized users through password verification.

This project focuses on designing and implementing a password-based door lock system using the STM32F103C8T6 Blue Pill microcontroller. The system uses a 4×4 keypad to enter the password, a 16×2 LCD to display status messages, and a relay module to control a solenoid door lock. The microcontroller compares the entered password with the predefined password stored in the program. If the password is correct, the relay is activated for a specified period, energizing the solenoid lock and allowing the door to open. If the password is incorrect, the door remains locked, and an error message is displayed on the LCD.

The hardware setup consists of the STM32 microcontroller, a 4×4 keypad, a 16×2 $I^2C$ LCD, a relay module, a solenoid door lock, and suitable power supplies. The GPIO pins of the microcontroller are used to interface with the keypad and relay, while the $I^2C$ communication interface is used to connect the LCD. The system is programmed using the Arduino IDE, and the program is uploaded to the STM32 microcontroller using a suitable programmer or bootloader.

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# CHAPTER 1
# INTRODUCTION

Security is an important requirement in homes, offices, educational institutions, laboratories, and other places where valuable items and personal belongings are stored. Traditional door-locking systems use physical keys to lock and unlock doors. However, keys can be lost, misplaced, or duplicated, which may create security concerns. To overcome these limitations, electronic locking systems provide an alternative method of controlling access. A password-protected door lock allows users to unlock a door by entering a predefined password instead of using a conventional physical key.

The project "Password-Protected Door Lock Using STM32F103C8T6 Blue Pill" aims to develop a simple, reliable, and cost-effective electronic security system. The STM32F103C8T6 Blue Pill microcontroller acts as the main control unit and manages the operation of all connected components. The system consists of a 4×4 matrix keypad, a 16×2 LCD display with an I2C interface, a buzzer, a relay module, and a solenoid door lock. These components work together to provide password-based access control.

The 4×4 matrix keypad is used to enter the password. The microcontroller reads the entered characters and compares them with the predefined password stored in the program. Based on the result, the system determines whether access should be granted or denied. The 16×2 LCD displays messages such as "Enter Password," "Access Granted," and "Wrong Password," helping the user understand the current status of the system. The I2C interface simplifies the connection between the LCD and the microcontroller by reducing the number of required signal connections.

When the correct password is entered, the microcontroller activates the relay module. The relay acts as an electrically controlled switch and controls the power supplied to the solenoid lock, allowing the locking mechanism to unlock according to its design. When an incorrect password is entered, the system denies access, displays an error message, and can activate the buzzer to provide an audible alert. These features improve user interaction and help prevent unauthorized access through the normal password-entry process.

The software is developed using the Arduino IDE with the appropriate STM32 board support package and compatible libraries. The program handles keypad input, password verification, LCD communication, buzzer alerts, and relay control. The hardware and software are integrated and tested to verify the response of the system to correct and incorrect password entries. Proper electrical connections and suitable power supplies are necessary to

ensure safe and reliable operation of the relay and solenoid lock.

The main objective of this project is to develop an affordable and user-friendly electronic door-locking system while gaining practical knowledge of embedded systems and microcontroller programming. The project demonstrates important concepts such as GPIO interfacing, keypad scanning, I2C communication, digital output control, and relay-based switching. It can serve as a basic prototype for access-control applications in homes, small offices, and laboratories. In the future, additional features such as password changing, automatic relocking, restrictions on repeated incorrect attempts, and alternative authentication methods can be introduced to improve the system's security and functionality. Overall, the project demonstrates how a microcontroller-based system can provide a practical alternative to conventional key-based door locks.

# CHAPTER 2
# REQUIREMENT STUDY AND ANALYSIS

Requirement study and analysis is an important stage in the development of an embedded system. It involves identifying the purpose of the project, understanding the problems to be solved, and determining the hardware and software resources required for implementation. In this project, the study focuses on developing a password-protected door locking system using the STM32F103C8T6 Blue Pill microcontroller. The requirements are analyzed to ensure that the system can accept password input, verify the password, display the system status, provide alerts for incorrect passwords, and control the solenoid lock through a relay module. Proper identification of these requirements helps in designing, implementing, and testing the system effectively.

## 2.1 Problem Definition

Traditional door locking systems generally depend on physical keys to provide access to homes, offices, laboratories, and other restricted areas. Physical keys can be lost, misplaced, or duplicated, which may create security concerns and inconvenience for users. Managing access using conventional locks can also be difficult when several people need to use the same door. Therefore, an alternative system is required to provide controlled access through an electronic method.

The problem addressed in this project is the development of a password-protected door lock that allows access only when the correct password is entered. The system must receive input from the user, verify the password, and operate the locking mechanism according to the result. It should also inform the user whether the entered password is correct or incorrect. The STM32F103C8T6 microcontroller is used to process the input and control the connected components. A keypad is used for password entry, an LCD displays messages, a buzzer provides an alert, and a relay module controls the solenoid lock. The main aim is to develop a simple, affordable, and practical electronic security prototype that demonstrates password-based access control.

## 2.2 Proposed System

The proposed system is a password-protected electronic door lock developed using the STM32 Blue Pill microcontroller. It uses a 4×4 matrix keypad as the input device, allowing users

to enter a predefined password. The microcontroller processes the entered characters and compares them with the password stored in the program. A 16×2 LCD with an I2C interface provides information about the system operation, including password entry, access status, and error messages. This allows the user to understand the result without requiring a separate computer or monitoring device during normal operation.

When the correct password is entered, the microcontroller sends a control signal to the relay module. The relay switches the solenoid lock circuit, allowing the door to unlock. When an incorrect password is entered, the system denies access, displays an appropriate error message on the LCD, and activates the buzzer to alert the user. The microcontroller coordinates these operations through its programmed control logic. The proposed system reduces dependence on physical keys and demonstrates how embedded systems can be used in electronic security applications. The design can also be improved in the future by adding password-changing facilities, multiple-user authentication, fingerprint recognition, or remote monitoring.

## 2.3 System Requirements

System requirements describe the functions and operating conditions necessary for the successful implementation of the project. The primary functional requirement is that the system must accept password input through the keypad and verify it using the predefined password stored in the program. The system must display suitable messages on the LCD according to the result of verification. It should activate the relay module when the password is correct and provide an alert through the buzzer when an incorrect password is entered. The control program must coordinate the keypad, display, buzzer, relay, and locking mechanism so that the system responds appropriately to user input.

The system must also satisfy basic non-functional requirements such as reliability, simplicity, affordability, and ease of operation. The keypad should respond correctly to key presses, the LCD should display readable messages, and the relay should operate according to the programmed logic. The components must receive suitable power, and the electrical connections must be stable. The system should be easy to test and maintain, with the password and control behavior defined clearly in the program. These requirements help ensure that the prototype performs its intended functions consistently under normal operating conditions.

### 2.3.1 Hardware Requirements

The hardware requirements include all the physical components necessary to construct and operate the password-protected door lock system. The STM32F103C8T6 Blue Pill development board acts as the main controller. It receives keypad input, verifies the password, and controls the output devices using its GPIO pins. The 4×4 matrix keypad allows the user to enter the password, while the 16×2 LCD with an I2C backpack displays system instructions and verification results. The I2C interface simplifies the display connections by using data and clock lines for communication with the microcontroller.

A buzzer is included to produce an audible alert when an incorrect password is entered or when an alert condition is detected by the program. The relay module acts as an electrically controlled switch and allows the microcontroller to control the solenoid lock circuit. The solenoid lock performs the physical locking and unlocking operation based on the power supplied through the relay contacts. A 3.7 V battery and a suitable 5 V boost converter are used to provide power to compatible components, while a separate 9 V battery is intended for the relay and solenoid circuit if their electrical ratings permit. A breadboard and jumper wires are used to establish the connections between the components. All power supplies, relay contacts, and the solenoid lock must be selected and connected according to their voltage and current requirements to ensure safe operation.

### 2.3.2 Software Requirements

The software requirements include the programming environment, board support, libraries, and firmware needed to control the hardware components. The Arduino IDE can be used to write, compile, and upload the program to the STM32F103C8T6 Blue Pill when a compatible STM32 Arduino core and board configuration are installed. Embedded C/C++ programming is used to implement the main control logic, including keypad input handling, password verification, LCD communication, buzzer activation, and relay switching. The program defines how the system responds to correct and incorrect passwords and determines which messages are displayed to the user.

Compatible libraries can simplify communication with the keypad and LCD modules. The keypad library helps detect the keys pressed by the user, while the LCD library supports displaying text through the I2C interface. A compatible upload method, such as an ST-Link programmer or another supported programming setup, is required to transfer the compiled program to the microcontroller. Debugging and testing help identify errors in the program

and verify that the hardware responds correctly. The software must be developed and tested carefully so that the relay operates according to the password verification result and the system performs its intended locking and unlocking functions.

### 2.3.3 Other Requirements

In addition to hardware and software, the project requires proper circuit assembly, suitable power management, and systematic testing. Basic knowledge of embedded programming, STM32 GPIO configuration, keypad interfacing, I2C communication, relay operation, and electronic locking mechanisms is useful for implementing the system. The components must be connected according to their pin configurations, and the operating voltage of each module must be checked before applying power. The solenoid lock should not be powered directly from the microcontroller because it requires more current than a GPIO pin can provide. Instead, a suitable relay or driver circuit must be used, with appropriate protection against voltage spikes generated by the solenoid.

The system should be tested under different conditions to confirm that the password verification process, LCD messages, buzzer alerts, relay switching, and locking mechanism work as expected. Testing should include both correct and incorrect passwords and verification that the lock returns to its intended state after access is granted. The power supplies must provide sufficient current for the connected devices, and the grounds and switching connections must follow the circuit design and module requirements. These additional requirements are important for improving the reliability and safety of the prototype. Proper testing and troubleshooting also help identify wiring errors, incorrect program settings, and component compatibility issues before the system is used in a practical demonstration.

# CHAPTER 3
# SYSTEM DESIGN AND DEVELOPMENT

The system design and development of the Password-Protected Door Lock Using STM32 Microcontroller focuses on integrating hardware and software components to provide secure access control. The STM32F103C8T6 Blue Pill acts as the main controller, processing keypad input and controlling the door-lock mechanism based on password verification.

The system consists of a 4×4 matrix keypad, a 16×2 LCD with an $I^2C$ interface, a relay module, a solenoid door lock, a buzzer, and a suitable power supply. The ST-Link V2 is used for programming and debugging the microcontroller.

The complete system is designed to provide a simple, low-cost, and effective security solution for homes, offices, laboratories, and other restricted areas. The modular design also allows future improvements, such as password changing, limited login attempts, and additional security alerts.

## 3.1 System Architecture

The system architecture describes the major components of the password-protected door lock system and the relationship between them. The STM32F103C8T6 Blue Pill microcontroller acts as the central processing and control unit. It receives password input from the 4×4 matrix keypad, verifies the entered password, and controls the LCD display, buzzer, and relay module.

The 4×4 matrix keypad is used to enter the password. The 16×2 LCD with an I2C module displays messages such as “Enter Password,” “Access Granted,” and “Access Denied.” The buzzer provides an audio indication during system operation.

When the correct password is entered, the STM32 activates the relay module, which controls the solenoid door lock. When the password is incorrect, the system displays an access-denied message and activates the buzzer as programmed.

The system uses a 3.7 V battery connected to a 5 V boost converter to supply power to the STM32 board and LCD. A separate 9 V battery supplies power to the relay module and solenoid lock. The two battery supplies are separate.

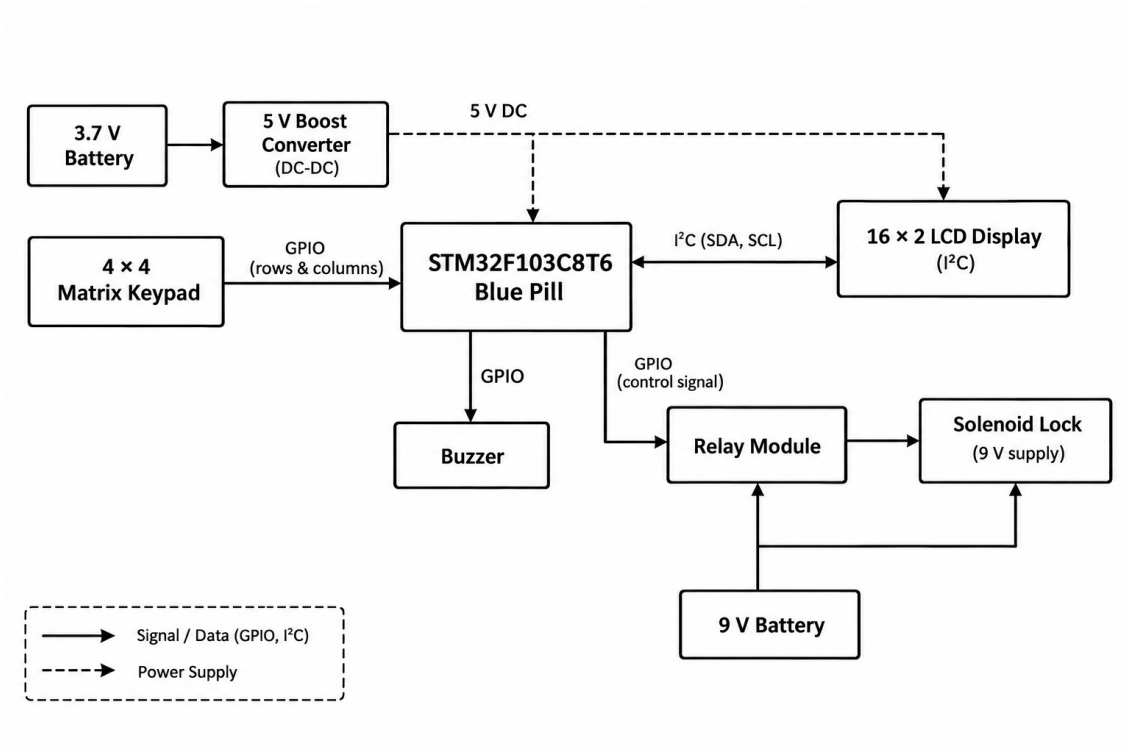



Figure 1: Block Diagram of the Password-Protected Door Lock System

### 3.1.1 The Architecture Consists of the Following Major Components

1. **Programming Environment:** The Arduino IDE is used to write, compile, and upload the control program to the STM32 board.

2. **STM32 Microcontroller:** The STM32F103C8T6 Blue Pill processes the entered password and controls the connected devices.

3. **4×4 Matrix Keypad:** The keypad allows the user to enter the password for door access.

4. **16×2 LCD Display:** The LCD displays password prompts, access status, and other system messages through the I2C interface.

5. **Buzzer:** The buzzer provides audible feedback for incorrect password entries and other programmed alerts.

6. **Relay Module:** The relay acts as an electrically controlled switch to control the power supplied to the solenoid lock.

7. **Solenoid Door Lock:** The solenoid lock provides the physical locking mechanism and changes its state according to the relay-controlled power circuit.

8. **Power Supply:** A 3.7 V battery with a boost converter provides a 5 V supply for compatible components, including the controller and LCD. A separate 9 V battery supplies the relay and solenoid lock, provided that their voltage and current ratings are suitable.

9. **Circuit Diagram**

   The circuit diagram illustrates the electrical connections between the STM32F103C8T6 Blue Pill microcontroller and the peripheral components, including the 4×4 matrix keypad, 16×2 LCD display with I2C module, buzzer, relay module, and solenoid door lock.

   The keypad is connected to the GPIO pins of the STM32 to enter the password. The LCD communicates through the I2C interface and displays the system status. The buzzer provides audio feedback, while the relay module controls the power supply to the solenoid lock.

   The system uses a 3.7 V battery connected to a 5 V boost converter to power the STM32 board and LCD. A separate 9 V battery supplies the relay module and solenoid lock, subject to their voltage and current ratings.

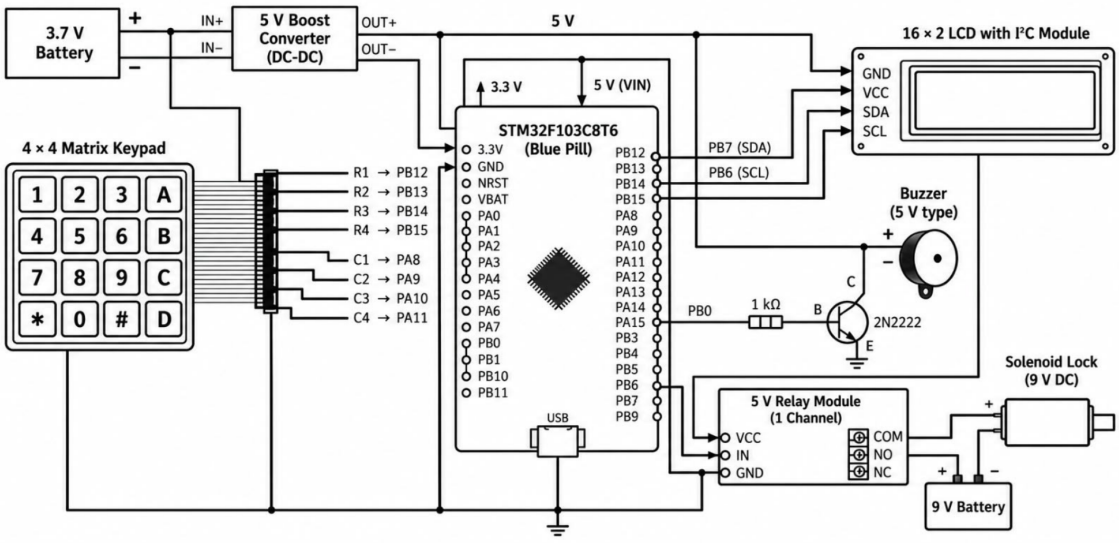



Figure 2: Circuit Diagram of the Password-Protected Door Lock System

10. **Pin Connection**

The STM32F103C8T6 Blue Pill microcontroller is connected to the keypad, LCD display, buzzer, and relay module through its GPIO pins. Table 1 shows the pin connections used in the system. These connections should match the actual hardware wiring and uploaded program.

- **Keypad:** PA0–PA3 are connected to the four row pins, while PA4–PA7 are connected to the four column pins. The STM32 scans these pins to detect key presses.
- **LCD Display:** SDA carries data and SCL carries the clock signal for I2C communication. PB7 and PB6 are commonly used for SDA and SCL, respectively, with the STM32F103C8T6 Arduino configuration.
- **Buzzer:** The buzzer is controlled through a GPIO output pin to provide audio feedback.
- **Relay Module:** The relay input receives a control signal from the STM32. The relay contacts switch the solenoid lock's power supply.
- **Power Supply:** The 3.7 V battery feeds the 5 V boost converter for the STM32 and LCD. The separate 9 V battery powers the relay and solenoid only if their ratings support it.

Table 1: Pin Connections of the System

| **Component** | **Component Pin** | **STM32 Connection** |
|---|---|---|
| Keypad | Row 1 (R1) | PA0 |
| Keypad | Row 2 (R2) | PA1 |
| Keypad | Row 3 (R3) | PA2 |
| Keypad | Row 4 (R4) | PA3 |
| Keypad | Column 1 (C1) | PA4 |
| Keypad | Column 2 (C2) | PA5 |
| Keypad | Column 3 (C3) | PA6 |
| Keypad | Column 4 (C4) | PA7 |
| LCD I2C | SDA | PB7 |
| LCD I2C | SCL | PB6 |
| Buzzer | Signal / IN | PB1 (verify) |
| Relay Module | IN / Signal | Verify against code |
| LCD I2C | VCC and GND | Suitable supply and GND |
| Relay Module | VCC and GND | According to module rating |
| Solenoid Lock | Power terminals | Through relay contacts |

## 3.2 Data Flow Diagram

The Data Flow Diagram illustrates the flow of data in the password-protected door lock system. The user enters a password through the keypad. The STM32 microcontroller checks the entered password and controls the LCD, buzzer, and relay module accordingly. The relay module controls the solenoid lock.

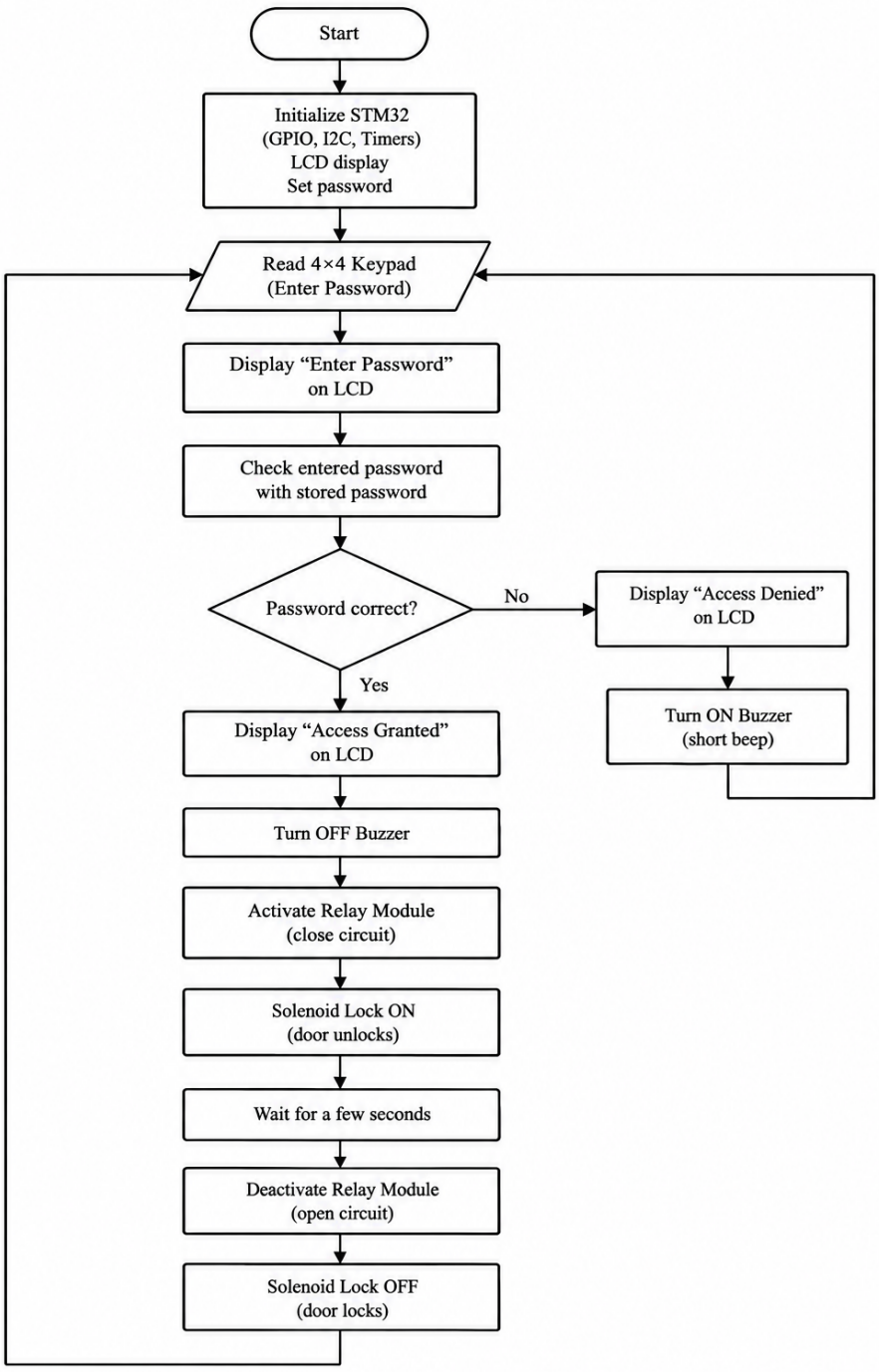



Figure 3: Flow Chart of the Password-Protected Door Lock System

# CHAPTER 4
# RESULTS AND DISCUSSION

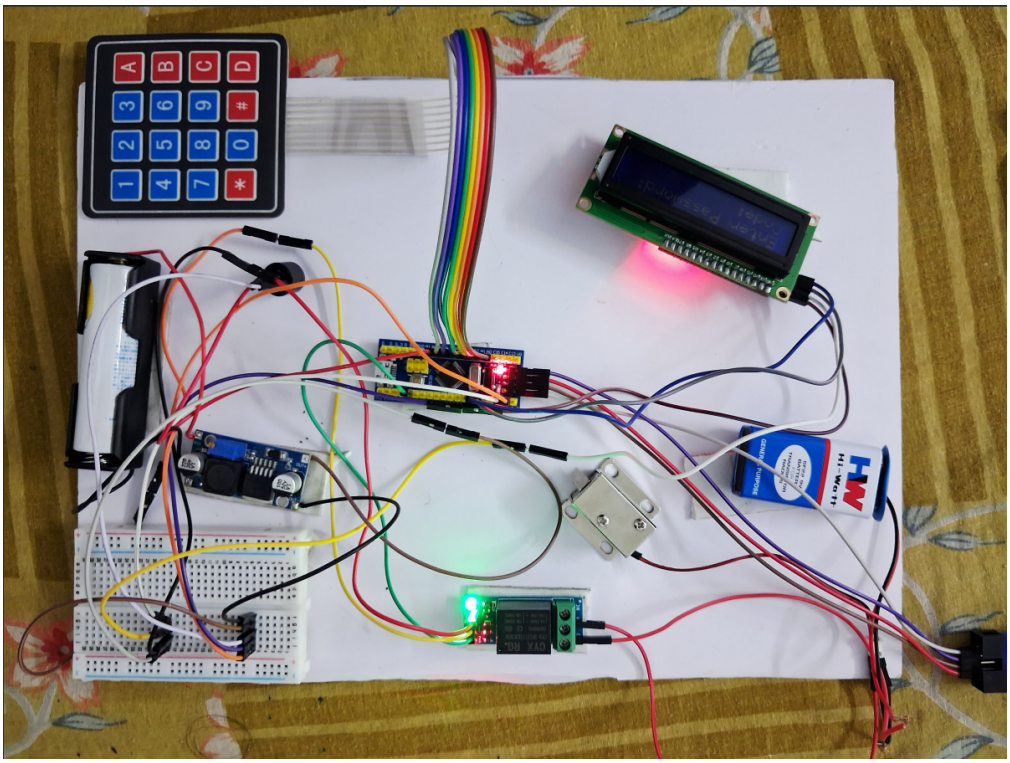



Figure 4: Output

The output of the Password-Protected Door Lock Using STM32F103C8T6 Blue Pill demonstrates the operation of the microcontroller along with the 4×4 matrix keypad, 16×2 LCD display, buzzer, relay module, and solenoid lock. The system provides password-based access control by accepting the password entered by the user and verifying it against the predefined password stored in the program. The STM32 microcontroller acts as the central control unit and processes the input received from the keypad to determine whether access should be granted or denied.

When the user enters a password through the keypad, the microcontroller reads the input and performs the password verification process. The LCD displays suitable messages to guide the user and indicate the authentication result. When the correct password is entered, the system displays an access-granted message and activates the relay module. The relay switches the power supplied to the solenoid lock, allowing the locking mechanism to unlock according to its design. This enables authorized users to access the secured area without

using a conventional physical key.

When an incorrect password is entered, the system denies access and displays an error message on the LCD. The buzzer can also produce an audible alert to indicate an incorrect password entry. The relay remains in the appropriate state to prevent the unlocking operation. The combination of the keypad, LCD, buzzer, relay module, and solenoid lock allows the system to perform password verification, provide user feedback, and control the door-locking mechanism.

Overall, the system integrates password authentication and electronic lock control into a single embedded application. It demonstrates the practical use of the STM32F103C8T6 microcontroller in security-related applications and provides experience in keypad interfacing, I2C communication, digital input and output control, and relay operation. The project offers a simple and cost-effective prototype for basic electronic access control. Its reliability and security depend on the correct implementation of the password verification logic, suitable electrical connections, and proper operation of the relay and solenoid lock.

# CHAPTER 5
# CONCLUSION AND FUTURE SCOPE

The Password-Protected Door Lock Using STM32F103C8T6 Blue Pill is designed to provide a simple and cost-effective electronic access-control system. The project demonstrates how a microcontroller can be used to verify a password and control a physical door-locking mechanism. It combines hardware components such as a 4×4 matrix keypad, 16×2 LCD display with an I2C interface, buzzer, relay module, and solenoid lock with embedded software developed using the Arduino IDE. The main aim of the project is to reduce dependence on conventional physical keys and provide a basic method of controlling door access through password authentication. This chapter presents the overall summary of the project, its significance, and the possible improvements that can be introduced in future versions.

## 5.1 Overall Summary of the Project

The project focuses on the design and development of a password-based electronic door-locking system using the STM32F103C8T6 Blue Pill microcontroller. The microcontroller acts as the central control unit and manages communication between the input and output components. The user enters a password through the 4×4 matrix keypad, and the system compares the entered password with the predefined password stored in the program. A 16×2 LCD displays instructions and authentication results, helping the user understand the current status of the system. When the correct password is entered, the microcontroller activates the relay module, which controls the power supplied to the solenoid lock and enables the unlocking operation. If an incorrect password is entered, the system displays an error message and can activate the buzzer to provide an audible alert. The integration of these components demonstrates the practical application of embedded systems, digital input and output, I2C communication, and electrical load control. The project also provides useful experience in circuit assembly, microcontroller programming, component interfacing, and troubleshooting. Overall, the system provides a basic and economical prototype for password-based access control, although its practical security depends on correct implementation, reliable power management, and suitable protection of the hardware and password.

## 5.2 Future Scope

The project can be improved in several ways to enhance its security, convenience, and functionality. One possible improvement is to introduce a password-changing feature so that authorized users can update the password without modifying the program. A limit on consecutive incorrect password attempts can also be implemented to reduce repeated unauthorized access attempts. An automatic relocking feature can help ensure that the door returns to its secure state after a predefined period. The system could further be enhanced by adding an RFID card reader, fingerprint sensor, or other authentication method to provide additional security. Wi-Fi or Bluetooth connectivity could be incorporated to enable remote monitoring, access notifications, and control through a mobile application. An alarm system could also be added to notify users when repeated incorrect passwords are entered or when unauthorized access is suspected. In addition, a backup power supply could help maintain operation during power interruptions. Future development may also include improved password protection, a more compact circuit design, and a stronger enclosure for the electronic components. These enhancements could make the system more suitable for practical applications in homes, offices, laboratories, and other environments requiring controlled access.